

CALL BY VALUE & CALL BY REFERENCE

Agenda

- Call by value
- Call by reference
- References

Methods of calling function

- Call by value
- Call by reference

Call by value

- Call by Value, means that a copy of the data is made and stored by way of the name of the parameter.
- i.e., It copies the value of actual parameters into formal parameters.
- During execution whatever changes are made in the formal parameters are not reflected back in the actual parameters.
- i.e., Any changes to the parameter have **NO** affect on data in the calling function.

PROGRAM

```
#include<stdio.h>
#include<conio.h>
void swap(int,int) ;
void main()
{
    int a,b;
    printf("Enter the two integers\n");
    scanf("%d%d",&a,&b) ;
    printf("a=%d and b=%d before calling function\n",a,b) ;
    swap(a,b) ;
    printf("a=%d and b=%d after calling function\n",a,b) ;
    getch() ;
}
void swap(int a, int b)
{
    int temp;
    temp=a;
    a=b;
    b=temp;
    printf("a=%d and b=%d in function\n",a,b) ;
}
```

OUTPUT

Enter the two integers

5 10

a=5 and b=10 before calling function

a=10 and b=5 in function

a=5 and b=10 after calling function

Call by Reference

- A reference parameter "refers" to the original data in the calling function.
- Thus any changes made to the parameter are ALSO MADE TO THE ORIGINAL variable.
- Reference(address) of the original variable is passed.
- Function does not create its own copy, it refers to the original values by reference.
- Functions works with the original data and changes are made in the original data.

PROGRAM

```
#include<stdio.h>
#include<conio.h>
void swap(int*,int*);
void main()
{
    int a,b;
    clrscr();
    printf("Enter the two integers\n");
    scanf("%d%d",&a,&b);
    printf("a=%d and b=%d before calling function\n",a,b);
    swap(&a,&b);
    printf("a=%d and b=%d after calling function\n",a,b);
    getch();
}
void swap(int* a, int* b)
{
    int temp;
    temp=*a;
    *a=*b;
    *b=temp;
}
```


OUTPUT

Enter the two integers

5 10

a=5 and b=10 before calling function

a=10 and b=5 in function

a=10 and b=5 after calling function

References

- <https://www.slideshare.net>

THANK YOU